

Dr. Rakesh Bhaskaran Nair

Principal Investigator & Early-Career Research Leader

School of Mechanical & Manufacturing Engineering, Dublin City University, Glasnevin, Dublin 9, Ireland

rakesh.nair@dcu.ie | +353 89 943 1136 | [rakeshbair.github.io/RakeshBNair](https://github.com/rakeshbair) | [Google Scholar](https://scholar.google.com/citations?user=...) | ORCID 0000-0003-2848-6391 | linkedin.com/in/rakeshbair

29	1,220+	18	€220k+
PUBLICATIONS	CITATIONS	H-INDEX	FUNDING (PI/CO-I)

EDUCATION

Ph.D., Mechanical Engineering 2020

Shiv Nadar University, India

M.Tech., Manufacturing Engineering 2014

Anna University, India

B.Tech., Mechanical Engineering 2009

University of Kerala, India

APPOINTMENTS

Principal Investigator & ECR Leader 2023–

Dublin City University, Ireland

Postdoctoral Fellow 2022–23

Concordia University, Canada

Postdoctoral Fellow 2020–22

University of Alberta, Canada

AWARDS & HONOURS

- 2025 Communication Leadership Award, I-Form (Research Ireland)
- 2023 Best Poster (1st Prize), Intl. Thermal Spray Conference
- 2021 Mitacs Accelerate Award (competitive), Canada
- 2020 Competitive Postdoctoral Fellowship, U. Alberta

EDITORIAL & SERVICE

- Lead Guest Editor — **npj Materials Degradation** (Nature Portfolio)
- Lead Guest Editor — **Discover Surfaces** (Springer Nature)
- Working Group Member — EU COST Actions CA23155 & CA22143
- Reviewer for 15+ international journals

TEACHING & TRAINING

- MSc lecturer, DCU (IR diagnostics; thermal spray). FHEA (submitted).
- Trained: L-PBF (Aconity3D / Autodesk Netfabb); cold spray (U. Alberta)
- Mentored 9+ postgraduate researchers (3+ PhD, 6+ MSc), two to co-authorship on 3+ Q1 papers

RESEARCH PROFILE

Materials engineer specialising in the design and deposition of thermal & cold spray coatings and in real-time materials diagnostics, targeting degradation of components in extreme environments across aerospace, wind-energy and marine systems.

RESEARCH FOCUS

Thermal & cold spray coatings · high-entropy alloys · degradation mechanisms · erosion-corrosion · high-temperature & CMAS corrosion · infrared pyrometry & emissivity · real-time diagnostics · additive manufacturing (L-PBF) · powder sustainability.

FUNDING (€220,000+ AS PI / CO-I)

- OptiSift** — Metal Powder Quality Intelligence Platform. Research Ireland ARC Hub. **€185,000** (PI, 2026–27)
- Sustainable-Fuel Tribology of Ni Superalloys. TTRF, Japan. \$25,000 (Co-I, 2023)
- Cold Spray HEA Coatings for Extreme Environments. Mitacs Accelerate. \$15,000 (PI, 2021)

SELECTED PUBLICATIONS

Spectral Emissivity Hysteresis in NiTi Powders. *Optics & Laser Technology*, 2026.
Nickel-Dependent Emissivity Calibration of NiTi Powders. *Measurement*, 2025.
CMAS Corrosion Attack on Thermally Sprayed TBCs: A Review. *npj Materials Degradation*, 2024.
HEA Coatings Deposited by Thermal Spraying: A Review. *Metals*, 2023.
Wear & Corrosion of Thermally Sprayed AlCoCrFeMo HEA Coatings. *Advanced Engineering Materials*, 2022.
Enhanced Wear Resistance of AlCoCrFeMo Coatings via Thermal Spray. *Surface & Coatings Technology*, 2024.

DESIGN & EXECUTION (BUILT FROM FIRST PRINCIPLES)

- Spectral imaging & customised thermal/emissivity measuring systems for L-PBF (2026)
 - High-speed liquid impingement (150 m/s) & slurry erosion-corrosion rigs (2017-18)
- Also contributed to a sub-scale abradable test rig for aerospace seals (CRIAQ consortium, Pratt & Whitney Canada, 2022).